

A comparative evaluation of view synthesis techniques utilizing Neural Radiance Fields (NeRF): video To 3D avatar

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Introduction

The NeRF framework offers several advantages over existing techniques. Firstly, it does not require extensive training on large-scale 3D datasets, as it can learn from a sparse set of input views. Secondly, it avoids the high storage costs associated with voxel-based representations when modeling complex scenes at high resolutions. Lastly, NeRF's continuous nature allows it to represent complex real-world geometry and appearance with remarkable accuracy. In this work, we introduce NeRF, a novel deep learning framework that represents 3D scenes as continuous volumetric functions. Unlike conventional methods, NeRF leverages the power of fully-connected deep networks to model the density and emitted radiance at each point in space, relative to the viewer's position and orientation. This results in a representation that is not only photorealistic but also capable of rendering novel views with high fidelity.

objective

While advantages of NeRF is obvious, there are many disadvantages about it. The main obstacle of carry out NeRF is the numerous photos and data it requires and the low speed. Consequently, our project is aimed at exploring and comparing two methods: using several images or using a video to construct the 3D avatar content which are both based on NeRF. We centered on the quality of the outcome 3D video, conducting the analysis of the relationship between loss and step.

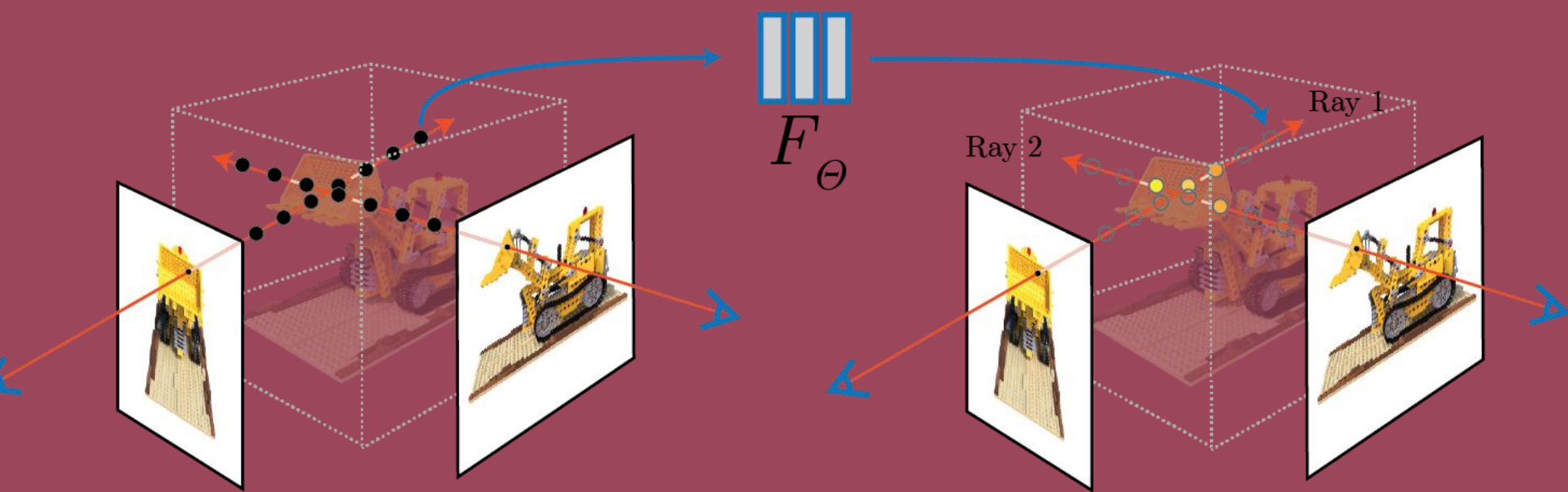
Methodology

Our framework operates in three stages:

Step 1: Define MLP to represent the scene

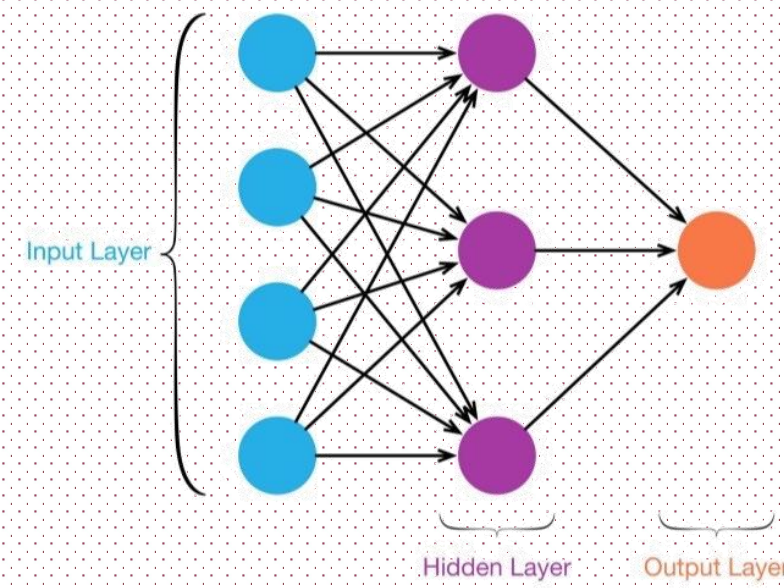
Step 2: Render scene from MLP using volumetric rendering

Step 3: Train MLP using the loss of the rendered scene



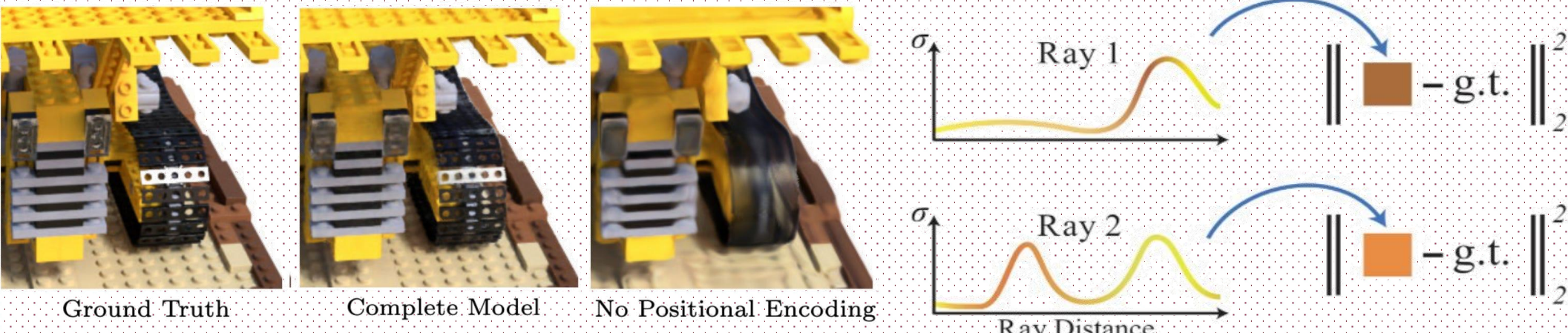
Basic thoery

- Multiple Fully-Connected Layers of Perceptrons
- Position coding
- Ray Casting



3D object and NeRf

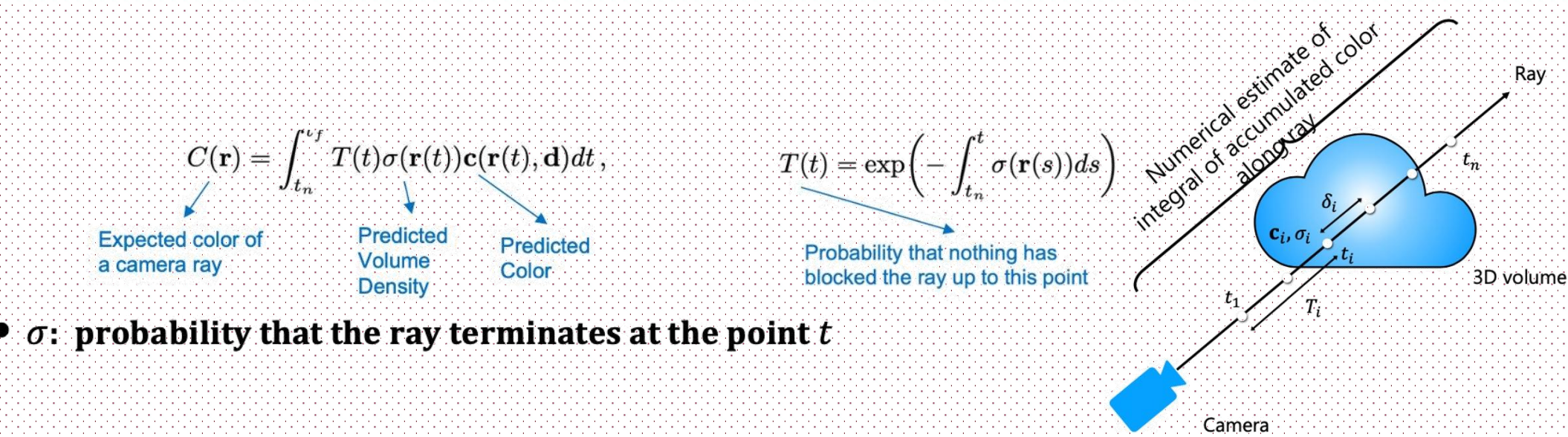
This process includes ray generation, ray casting, and volumetric rendering. During the ray generation phase, the system constructs a ray for each pixel, starting from the observer's position. In the ray casting phase, the system identifies the intersection points of the rays with the scene and performs lighting calculations, ultimately setting the calculated color as the pixel's color.



The architecture of NeRF includes a multilayer perceptron (MLP) that maps positions in 3D space to color and volume density. During training, the network optimizes its parameters through gradient descent to minimize the difference between the rendered images and the actual images.

Rendering

To render a scene, NeRF uses the classic technique of volume rendering, sampling along camera rays and projecting the output color and density into the image. We use the volume density to represent the probability that the ray will end up with an infinitesimal particle at some point in space. It is similar to differential transparency and controls the amount of accumulated radiation from the rays passing through the point. The directional emission radiation function is then used to represent the color of the ray emitted along the observed direction at a point in space. For each pixel in the virtual camera, samples are taken along the rays to generate a set of 3D points.



Each 3D point and the corresponding 2D observation direction are passed to the neural network as input to obtain the corresponding color and volume density. Use classic volume rendering techniques to add color and density to a 2D image. Then stratified sampling, using the trapezoidal rule to estimate the continuous integral. Hierarchical sampling is performed using a coarse network to generate initial sampling points and a fine network for finer sampling in key areas.

Results

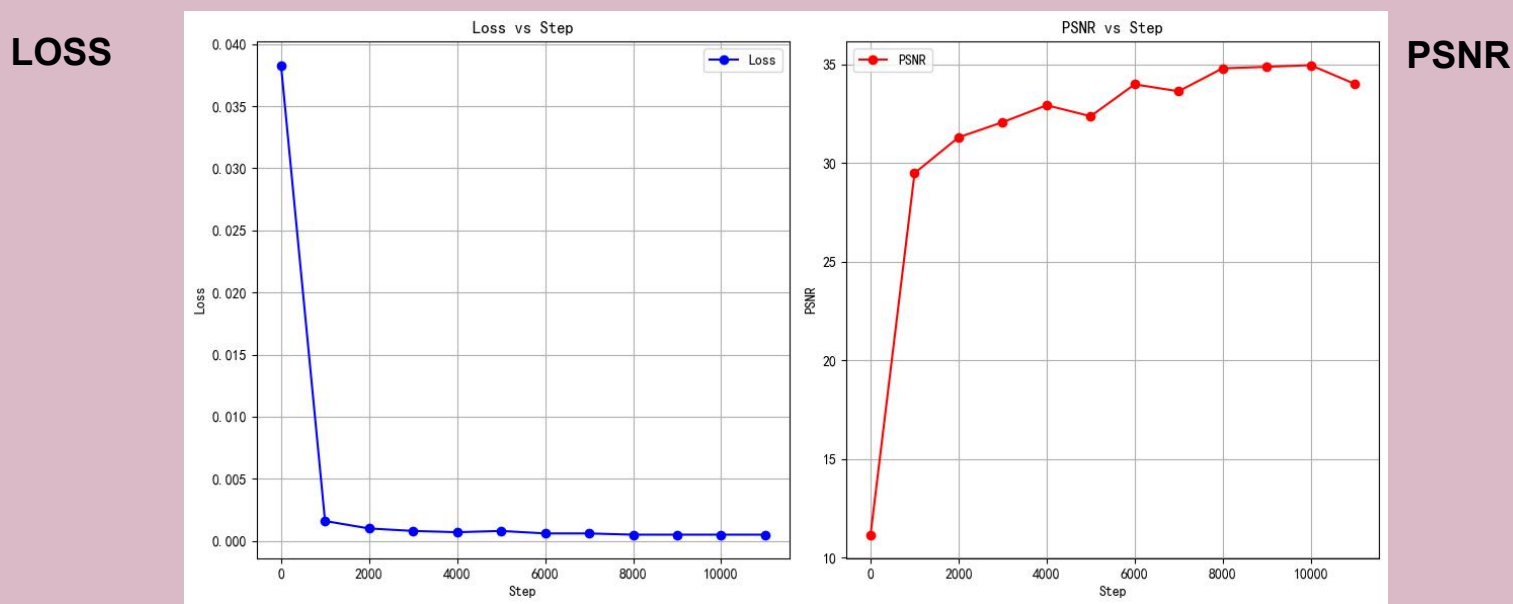
Programming environment:

Nan Huang	Yumo Chen
CPU:13代i7-13700H	CPU:12 i7-12700H
GPU:RTX 4060 8GB	GPU:NVIDIA GeForce RTX 3060 13.9Gb
RAM:16GB	RAM:15.7GB

Model:l1c/cv_nerf-3d-reconstruction-accelerate_damo

Experiment :

we use a video of our team member to began the frist tryout.As we can see,along with the step goes,the loss decrease sharply between step 0-2000; after step 2000,the data changes is gradual and approaches zero.

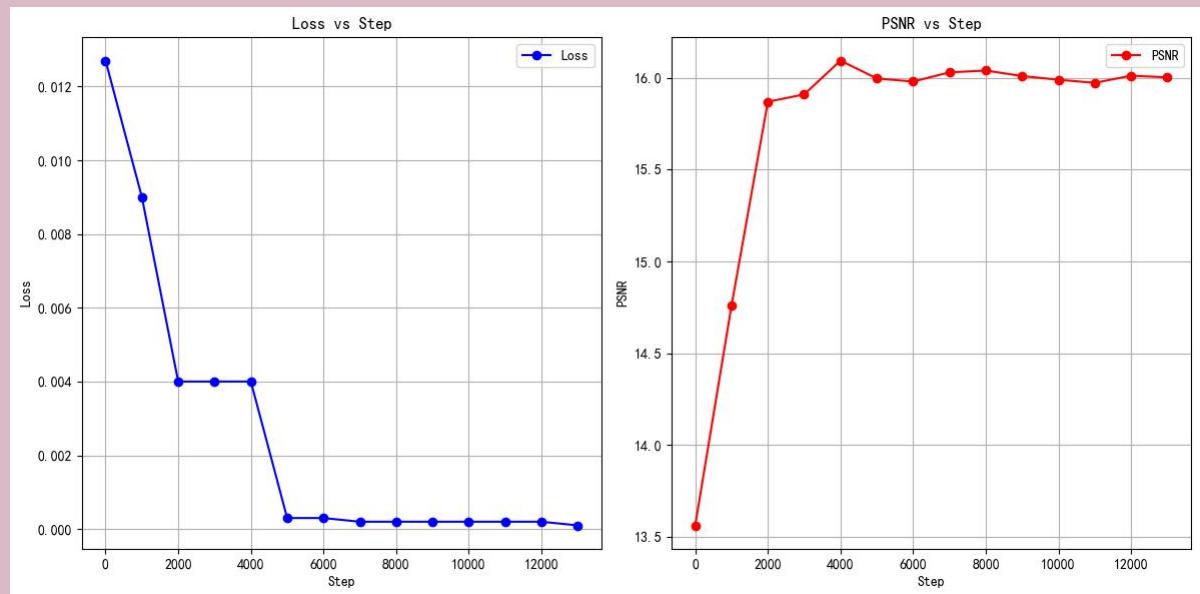


in the second tryout ,we record a 360° observation surrounding another teammate.(because our model is a bit shy so he refuse to show his face)



Comparing with the first experiment ,this avatar is much more three-dimensional.No matter the texture of the clothes or the sense of volume, the program seems smarter.

Obviously, the rate of loss is lower than before. And after we correct the value of PSNR, it raises along the step to form a better constructed 3D video with all perspective of the subjective from original video.



Conclusion

- NeRF ability to generate highly realistic and expressive talking portraits positions it as a powerful technological support for applications in virtual reality, metaverse, and digital humans, potentially driving technological advancements in these fields.
- Although NeRF may have slightly less impressive performance in terms of run time and model size compared to some existing technologies like ER-NeRF, its significant advancements in expressiveness and visual quality substantiate its potential and value in practical applications.
- The introduction of NeRF opens new avenues for future research on talking portrait synthesis on platforms with constrained computing capabilities, such as mobile devices, especially in the development of lightweight models.
- NeRF introduces an innovative Expression-Aware Module that learns expression information from target images, significantly enhancing the capture and reproduction of subtle facial expressions in the synthesis of realistic talking portraits.

Reference

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